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Simplifying Scientific Text

SPU_DeepSimplify



Complex text to clearer language

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The Problem: Why This Matters

Why scientific and medical text is hard for non-specialists

Aa

Complex Terminology

Specialized medical and scientific terms reduce accessibility.

Dense Sentence Structure

Long clauses and nested phrasing make reading difficult.

Statistical Overload

Numbers, percentages and evidence details can overwhelm readers.

Hallucination Risk

Naive simplification may remove, distort or add information.

Goal: produce simpler, faithful sentences without losing clinical meaning.

Input and Output Example

One complex sentence becomes a clearer sentence with the same meaning

Input

Complex scientific sentence

Available information was insufficient for reliable comparison of total hospital costs and postoperative pain/quality of life between the two approaches.

Simplify

Output

Simplified sentence

We did not have enough information to reliably compare total hospital costs, pain after surgery, or quality of life between the two approaches.

Simpler wording, preserved uncertainty

“Insufficient” becomes “not enough”. “Postoperative pain” becomes “pain after surgery”.

Illustrative rewrite of a development-data sentence. This is not a reported model prediction.

Evaluation Metrics

What each measure tells us, in simple terms

Metric	What it measures	How to read it
SARI	Quality of adding, deleting and keeping words, using source and references.	Higher score indicates better agreement with reference edits.
BLEU	Word-sequence overlap between the output and reference simplifications.	Higher means more overlap, not necessarily easier reading.
FKGL	Estimated US school grade needed to read the text.	Lower means a lower estimated reading grade.
Compression	Output length relative to source length.	Below 1 means shorter. Shorter alone does not mean better.
Sentence Split	How much the system divides the source into shorter sentences.	More splitting helps only when appropriate.
Levenshtein	Edit distance: insertions, deletions and substitutions needed to change the text.	Higher distance means more editing, not better quality.
Copies	How often the system repeats the original sentence.	More copying means fewer changes. Some sources need no edits.
Additions	Words introduced in the output compared with the source.	Additions may clarify a term or introduce unsupported content.
Deletions	Source words removed from the output.	Deletions may remove detail or lose important information.
Lexical Complexity	Estimated difficulty of the vocabulary.	Lower usually means simpler words. The scale depends on the scorer.

Edit statistics describe system behavior. Readability and meaning preservation must be assessed together.

Previous System and CLEF Result

45.00

Final SARI on Codabench

CLEF 2026 SimpleText
Task 1.1

Main pipeline: 44.91 SARI

After selective post-processing:
45.00 SARI

Stage	Published component
Terminology	MedSimplify source-term hints
Planning	Llama 3.1 8B produces a JSON plan
Strategy	Rules choose rephrase, split, delete or keep
Execution	Llama 3.3 70B generates the output
Output checks	Validation, fallback and selective revision

Both named models use inference without task-specific fine-tuning.

Previous System Workflow

Operation-aware two-stage simplification pipeline



Scientific Text Simplification Systems

Shared-task research comparison — Part 1 of 2

System / source	Input → output	Models and method	Reported results	Strengths and limitations	Readability
Writerslogic paper455.pdf	Sentence + 7 few-shot examples → 5 candidates → selected simplification. Full abstract → simplified document.	Claude Sonnet 4 generates sentence candidates at five temperatures; a rule-based selector rewards source-word retention, compression, simple vocabulary and plain-language terms, and penalizes exact copying. Documents: direct Llama 3.3 70B. GPT-4o-mini and Qwen3-8B explored.	Sentence SARI 47.43 Document SARI 43.96	Strength: diverse candidates and an explicit selector. Limit: surface criteria do not ensure meaning preservation; document outputs omit content; API cost and reproducibility concerns.	FKGL ↓ Sentence: 10.19 Document: 9.91
FOSU-YZH paper473.pdf	Sentence, with local context for some variants → 4 candidates → one output. Task 1.1 only.	DeepSeek-V4-Pro / DeepSeek-chat generate a base output, an alternative, a context-aware version and a cautious version. Filter invalid candidates; score compression, statistical cleanup and copying. Replace the base only with a clear gain. No fine-tuning; temperature 0.	Sentence SARI 47.01 (original) 46.23 (auto) No document run	Strength: conservative selection and attention to uncertainty. Limit: no independent semantic verification; small selector gain. The full four-candidate pipeline applies to the primary English subset.	FKGL ↓ — sentence 10.55 (original) 10.59 (auto)
ELiRF-UPV paper459.pdf	Sentences within document context → action labels → simplified sentences or document. Tasks 1.1 and 1.2.	RoBERTa-large + fusion heads + CRF predicts copy/delete/rephrase/split/merge actions; five-seed ensemble. Executors: Gemma 4 31B prompting, or BART / Flan-T5 with SFT/PEFT and action conditioning. Flan-T5 also uses DPO. Best official runs use Gemma without actions.	Official SARI Sentence 46.36 Document 46.22 Best: no action plan	Strength: explicit planning/generation separation; action and DPO experiments. Limit: planner gains are inconsistent; local training gains did not generalize well to official evaluation; no factuality guarantee.	FKGL ↓ Sentence: 7.87 Document: 7.77
AIIRLab paper457(1).pdf	Sentence + gist/action, or sentence alone → candidates → final rewrite. Abstract → simplified abstract.	AllenNLP semantic-role gist; Ministral 8B SFT and DPO; Gemma 4 31B few-shot alternative. Llama 3.1 70B combines three outputs with source/context and may rewrite. Document alternatives: Gemma agents, APIO, staged Llama 3.1 8B, and direct Llama baseline.	Sentence SARI 46.24 original 45.08 auto Document SARI 42.67 original 42.21 auto	Strength: compares SFT, DPO, prompting and ensemble generation. Limit: many calls; higher SARI need not mean lower FKGL. Document agents did not beat the baseline; LLM review is not a medical guarantee.	FKGL ↓ Sentence: 11.36 orig. 11.35 auto Document: 11.06 orig. 13.49 auto

Source: Combined_Research_Tables_RTL.pdf • Reported scores use different evaluation settings; they are not a single leaderboard.

FKGL = Flesch–Kincaid Grade Level; lower is easier.

Scientific Text Simplification Systems

Shared-task research comparison — Part 2 of 2

System / source	Input → output	Models and method	Reported results	Strengths and limitations	Readability
DUTH paper450.pdf	Sentence → simplified sentence. Document → direct output or assembled sentence outputs, with length-based fallback.	Gemma-2-9B-IT with grounded / EasySplit prompts; preserve numbers, comparisons and uncertainty; light post-processing. Document hybrid falls back to direct simplification when length is unsuitable. Candidate banks, selectors and other models explored. No simplification fine-tuning.	SARI Sentence 46.09 Document 46.16 Gemma-2-9B-IT	Strength: grounded instructions and a document fallback. Limit: instructions do not verify facts; lexical matching misses paraphrases; some thresholds are underspecified; individual stage contributions are not isolated.	FKGL ↓ Sentence: 11.75 Document: 11.75
NIL-UCM paper454.pdf	Sentence → direct simplification or copy / delete / rephrase / split. Document → plain-language summary.	Llama-3.1-8B, Mistral-7B and Gemma 2: role-based zero/few-shot, action classification, and LoRA on 849 abstract–PLS pairs (rank 64, 3 epochs). Document ensemble: four-model drafts → Llama selection → Gemini fact/readability review → Mistral revision, up to 3 rounds.	SARI Sentence 45.51 Document 45.34 Sentence best: Llama zero-shot	Strength: broad comparison of prompting, planning, training and agents. Limit: mostly SARI, no human evaluation; stronger agent models confound architecture effects; limited LoRA experiments were unsuccessful.	FKGL is mentioned as an evaluation metric. Numerical scores are not reported.
FOSU_cr paper453.pdf	15 numbered sentences per request, or an abstract + 3 real examples → simplification → statistical cleanup.	DeepSeek-V3 / deepseek-chat with three fixed Cochrane examples selected by length criteria from BioCLEAR references; not per-input retrieval. Temperature 0.3; document target 200–300 words. Regex removes statistics and evidence-certainty descriptions. No fine-tuning.	Sentence SARI 43.50 (local mean) No separate official sentence score Document SARI 48.85 orig. / 47.97 auto	Strength: simple few-shot pipeline with example-type ablations. Limit: removing certainty can lose medical meaning. Local evaluation references supply examples, which affects interpretation; no independent medical verification.	FKGL ↓ — document 8.80 (original) 8.54 (auto) No sentence FKGL value reported.
SPU_ DeepSimplify paper470.pdf	Sentence + MedSimplify hints + optional next sentence for planning → JSON plan → strategy → simplified output. Task 1.1 only.	MedSimplify terminology support → Llama 3.1 8B planner → rules selecting rephrase, split, delete or keep → Llama 3.3 70B executor → output checks/fallback. Selective post-processing revisits near-copy outputs with a Grade 8 target. Inference only; no fine-tuning.	Final SARI 45.00 No document run	Strength: traceable stages; smaller planner and larger executor; selective revision. Limit: planner/rule errors can select unsuitable actions; validity checks do not establish factuality; no component ablation and only a small post-processing gain.	FKGL ↓ Final system: 10.19

Source: Combined_Research_Tables_RTL.pdf • Reported scores use different evaluation settings; they are not a single leaderboard.

FKGL = Flesch–Kincaid Grade Level; lower is easier.

Additional Related Studies

Planning, jargon support, controllability and multi-agent simplification

Study	Data	Models and method	Reported results	Evaluation / caveat	Readability
Plan-Guided Text Simplification with Extended Contexts 2026 [1]	Newsela-auto 18,820 document pairs from 1,882 articles; multiple simplification levels.	Oracle alignment actions or pgdyn-plan predictions. BART-base fine-tuning vs Qwen2.5 7B/14B/32B and Gemma-3-27B prompting with 3 examples; process 1–5 paragraphs, then assemble.	SARI 51.5: Qwen-32B + Oracle, 1 paragraph. Predicted plans: 47.4 (1 paragraph); 47.0 (5). Action recall: 82% → 89.5%.	SARI, SMART, FKGL; CRF alignment and action micro-recall. Oracle and predicted-plan results must be distinguished.	FKGL ↓
JEBS 2025 [2]	PLABA 400 abstracts; 10,314 term spans; 21,595 manually annotated simplifications.	Detect jargon → classify substitution, explanation, generalization, exemplification or omission → generate. BERT/DeBERTa detectors/classifiers; Llama3-8B, Llama4 Scout and GPT-4o-mini generators.	Detection F1 0.4317 (DeBERTa). Classification mean F1 0.3588 (BERT-NER). GPT-4o-mini + 3 examples best overall generation.	Detection/classification F1; human ease, accuracy, completeness and concision (48 examples per action/version). No full-system SARI.	Human evaluation: simplicity / ease
Medical Text Simplification: Jargon Detection to Jargon-Aware Prompting 2025 [3]	MedReadMe + PLABA for detection. Simplification: 300 PLABA abstracts, 3,315 sentences.	Fine-tune BERT, RoBERTa, BioBERT and PubMedBERT for jargon detection. Llama-3.1-8B / Medicine-Llama3-8B simplify with direct prompts, detected/oracle jargon or oracle actions; assemble abstracts.	Best SARI 32.26; FKGL 15.36 (oracle actions). Direct prompt: SARI 29.87; FKGL 13.53. PLABA detection F1: 46.70%.	SARI, BLEU, BERTScore, FKGL; entity/token F1; human meaning, ease and fluency. Best SARI uses oracle information.	FKGL ↓ 15.36 (oracle actions) 13.53 (direct prompt)
Taming CATS 2026 [4]	Med-EASi, SimPA, WikiLarge, Newsela. Custom stratified 80/10/10 split and sampled data.	Instruction fine-tuning with separate control tokens for reading level or word/character length. Llama, Mistral and Qwen (1–14B); LoRA for larger models. No separate action planner.	SARI 67.78: Llama-13B, SimPA, FKGL control. Medical: 52.18, Mistral-7B, Dale-Chall control. Qwen3-1.7B medical: 52.09.	SARI, LENS, control MAE, BLEU, BERTScore, COMET. Not a CLEF test; higher SARI did not always improve LENS.	Readability controls: FKGL; Dale–Chall Target-control error: MAE
Let's Simplify Step by Step 2026 [5]	CEFR-SP (English). README++ in Arabic, English, French, Hindi and Russian.	CEFR transition matrix → dynamic-programming path → 3 high-STS examples → stepwise generation with history. Llama-3-8B, Gemma-2-9B, Mistral-7B prompting; trained mT5-base baseline.	README-EN best exact accuracy: 46.46% (Llama). Best adjacent accuracy: 96.46% (Gemma). Full Llama system: 37.37% exact; STS 69.90%.	Exact/adjacent accuracy, RMSE, Spearman, STS, BERTScore and human ratings. Planning cuts steps 22–42%; no SARI.	CEFR target level: Exact / Adjacent Accuracy; RMSE; Spearman correlation
Collaborative Document Simplification Using Multi-Agent Systems (AgentSimp) 2025 [6]	Newsela: 200 documents. GuoFeng: 100 English literary texts.	Document guidelines/structure → paragraph simplification and review → jargon/metaphor clarification → reconstruction and checking. GPT-3.5/GPT-4 roles; sequential/parallel communication and direct/incremental construction.	Newsela: SARI 41.58; D-SARI 30.72 (GPT-4, sequential + incremental). Direct GPT-4: 33.61. Trained Mistral-7B: 37.20 vs 33.73 untrained.	SARI, D-SARI, BARTScore, FKGL; human and LLM evaluation. Document results; review roles did not always raise SARI; API cost limited evaluation size.	FKGL ↓ Human evaluation

Source: Combined_Research_Tables_RTL.pdf • Reported scores use different evaluation settings; they are not a single leaderboard.

FKGL = Flesch–Kincaid Grade Level; lower is easier.

Related Work by Method

Prompting and Candidate Selection

Writerslogic

FOSU-YZH

DUTH

FOSU_cr

Fine-Tuning and Controlled Adaptation

ELiRF-UPV

NIL-UCM

Taming CATS

Planning and Structural Simplification

Plan-Guided Text Simplification with Extended Contexts

Let's Simplify Step by Step

SPU_DeepSimplify

Terminology-Supported Simplification

JEBS

Medical Text Simplification: From Jargon Detection to Jargon-Aware Prompting

Ensemble and Multi-Agent Simplification

AIIRLab

AgentSimp

Research Gap and Proposed Contribution



Research Gap

Higher benchmark scores do not always mean easier reading. The balance between simplification quality, meaning preservation, and inference cost needs further evaluation.



Proposed Contribution

Combine PEFT-based and planning-based candidates to improve readability and preserve meaning, while exploring selective use of large models to limit inference cost.

Proposed Ensemble System

PEFT-based branch

Train a small set of adapter weights with LoRA or QLoRA using paired examples. The adapted model generates Candidate A.

Planning-based branch

Run SPU_DeepSimplify to generate Candidate B. Possible refinements concern terminology, planning, rules, validation or selective revision.

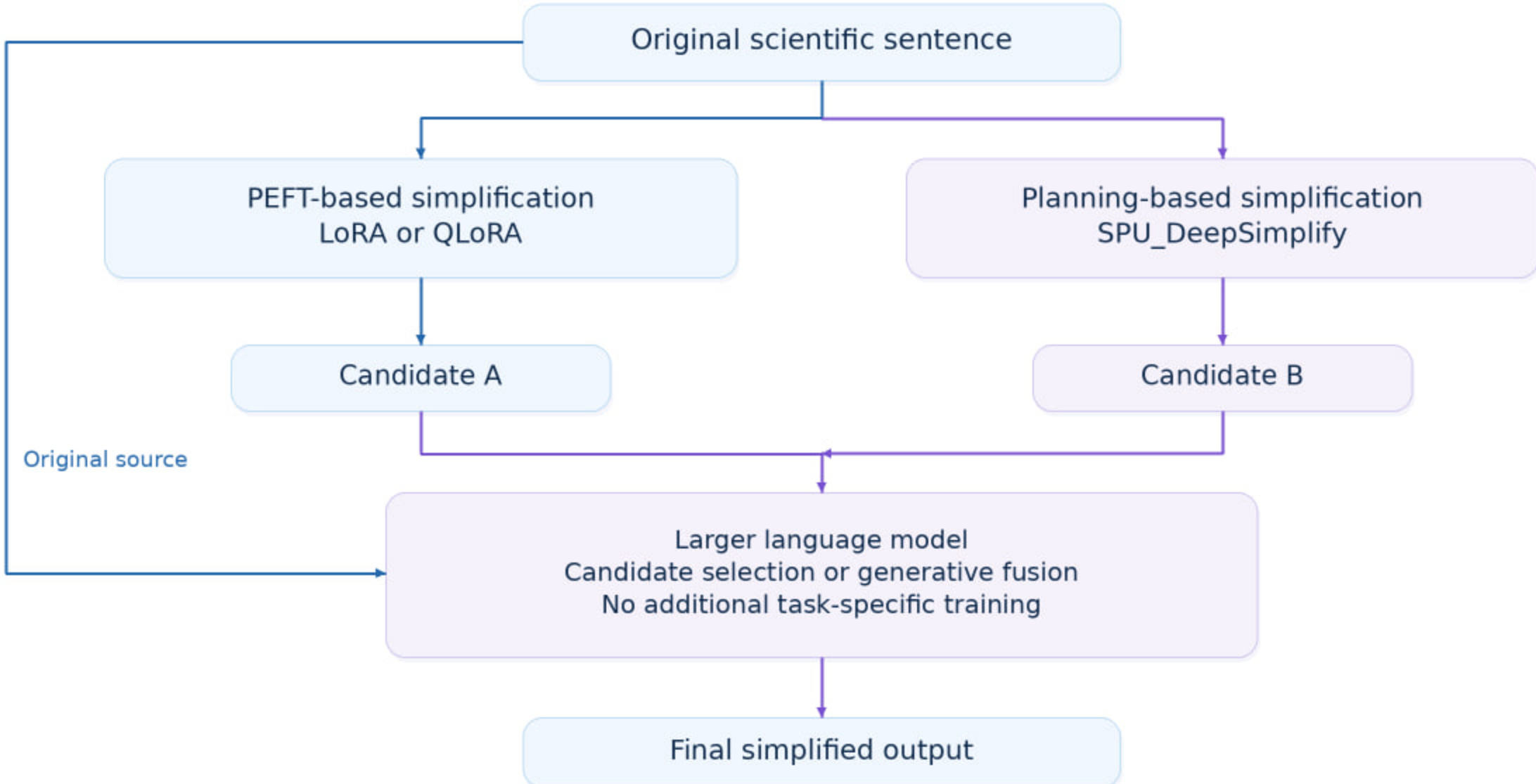
Final ensemble

A larger language model receives the source sentence and both candidates. It either selects one candidate unchanged or generates a new source-grounded simplification.

Compare each branch, any documented prior-system refinements and the final ensemble under the same evaluation conditions.

Proposed architecture; no new ensemble performance result is reported.

Proposed System Workflow



Development Data

Raw file counts before filtering and sequence-length exclusions

File	Records	Documents	Purpose
cochraneauto_sents_train.csv	11,510 sentences	849	Supervised training
cochraneauto_sents_val.csv	1,697 sentences	119	Validation and checkpoint selection
cochraneauto_sents_test.csv	1,512 sentences	117	Local test evaluation
cochraneauto_docs_train.csv	849 documents	849	Full training-document text

10 columns in each development file

Identifiers

pair_id — document ID
para_id — paragraph ID
sent_id — source sentence ID
simp_sent_id — aligned simplified sentence ID

Text pair

complex — original source
simple — reference simplification

Sentence rows hold sentences; document rows hold full texts.

Annotations and position

label — simplification operation
doc_pos — relative position
doc_quint — document fifth
doc_len — source sentence count

In document rows, sentence metadata are stored as lists. The 849 training documents match the sentence-training document IDs.

Document IDs do not overlap between the local training, validation and test splits.

Official Test Data

Sentence input: sentences_src.json

48,809 sentence records
1,734 distinct document IDs
6 fields; multiple collections and languages

Document input: documents_src.json

Full original document text per record
4 fields in the provided example
Complete-file record count: not verified

Main English subset: 3,679 sentences from 175 documents

Select source = "Cochrane-auto 2026" and language = "en"

Field	Sentence JSON	Document JSON
pair_id	Document identifier	Document identifier
source	Source collection	Source collection
language	Language code	Language code
complex	Original sentence	Full original document
para_id	Paragraph identifier	Not present in the example
sent_id	Sentence identifier	Not present in the example

With context: keep sentence IDs and match original document text by pair_id + source + language. The output remains sentence-level.

The full sentence JSON also contains earlier collections and local validation/test subsets. Reference simplifications are not model inputs.

Thank You